

Linking Transposase Domestication to Patterns of Eukaryotic Evolution

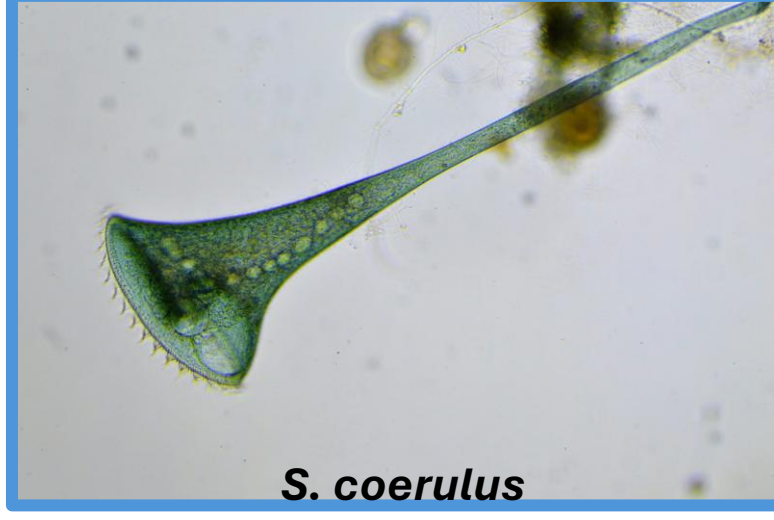
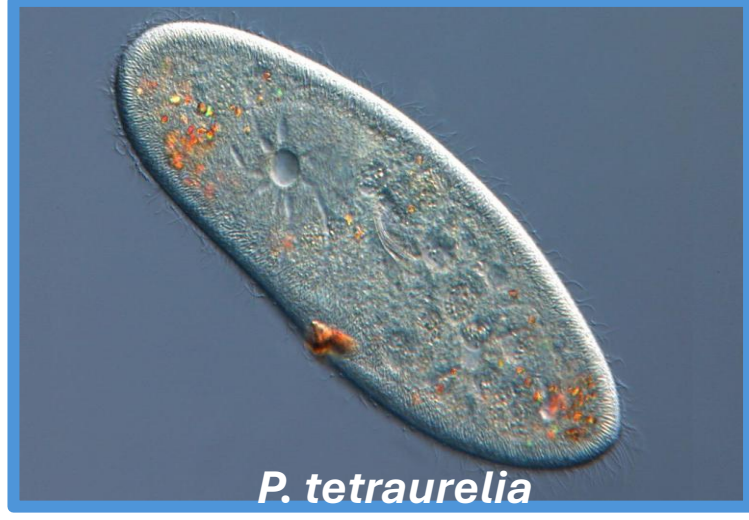
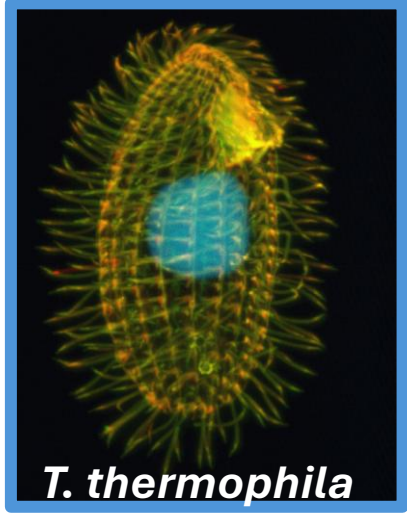


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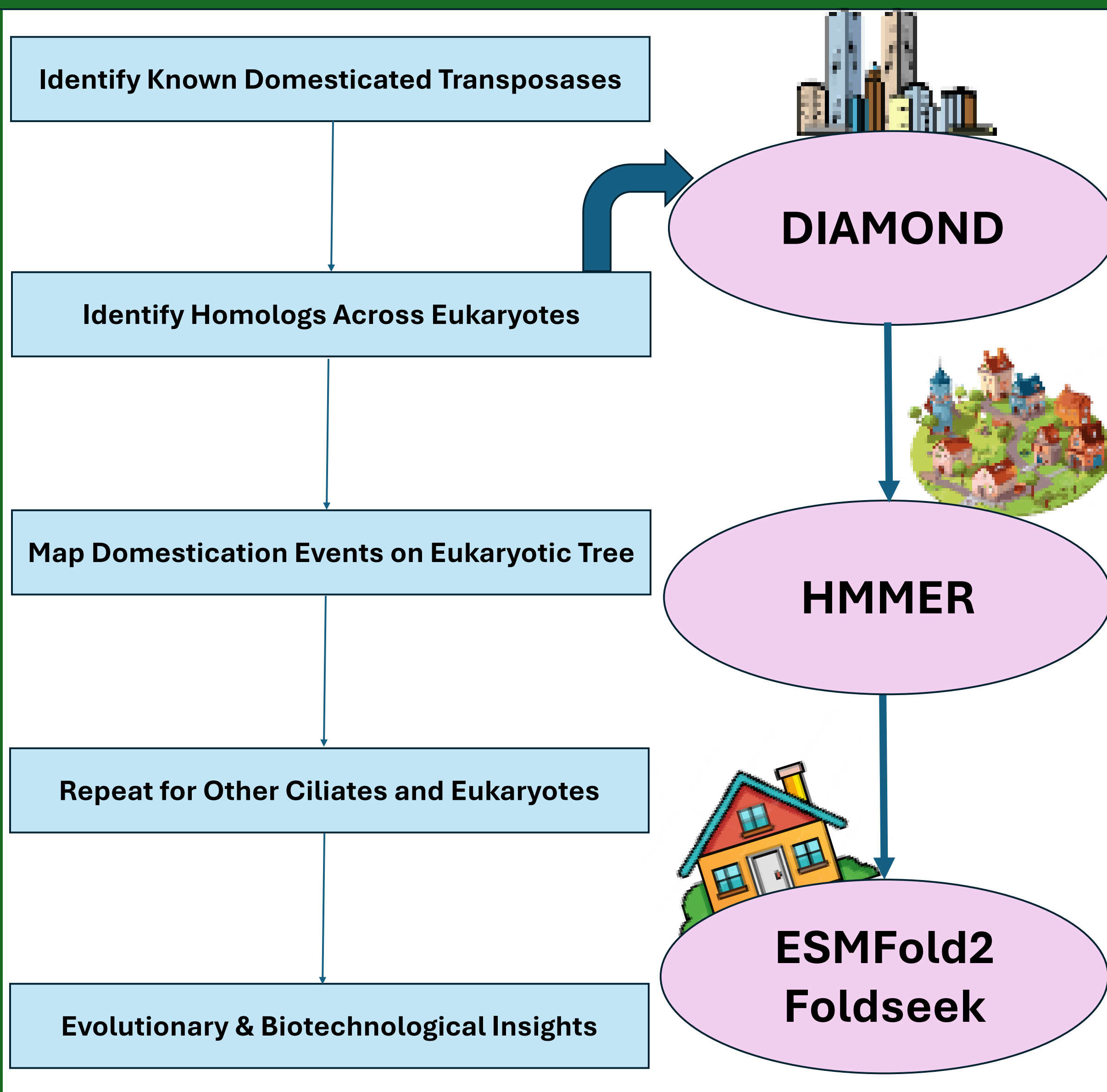
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Introduction

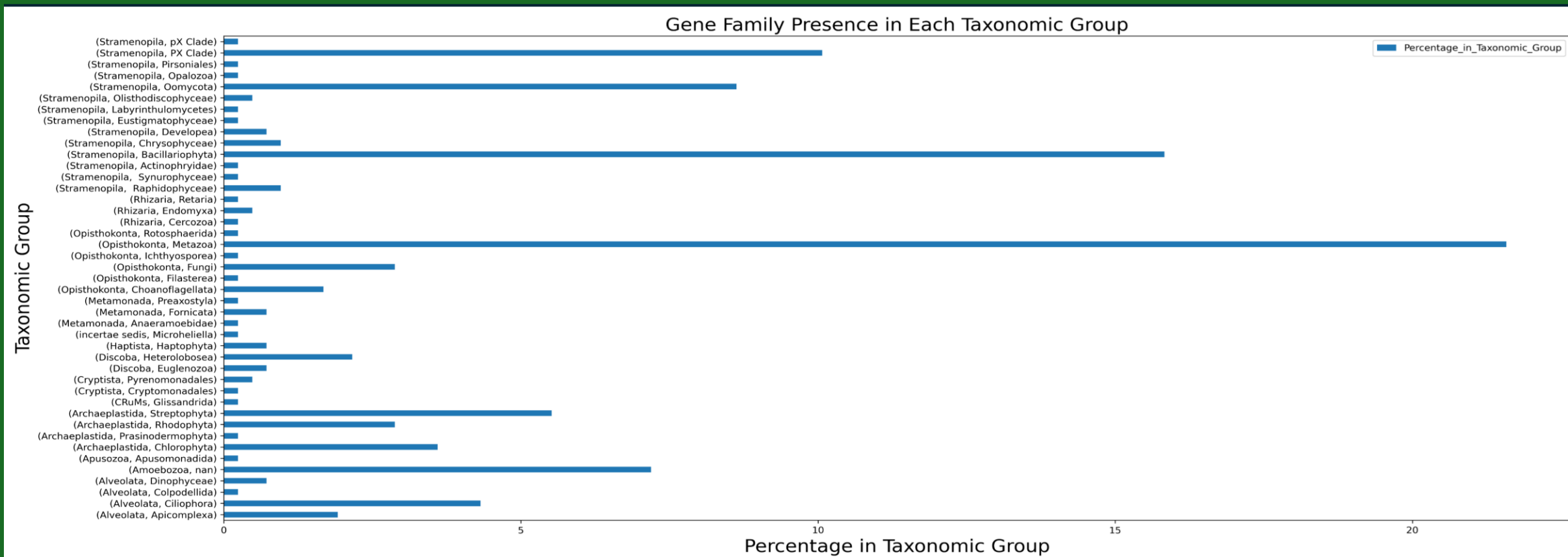
- Tetrahymena thermophila*, *Paramecium teraurelia*, and *Stentor coeruleus* are ciliate species.
- Focused on *T. thermophila* and *P. tetraurelia* because they are well characterized and understood → Diverged during the Cambrian Explosion and belong to the class Oligohymenophorea.
- Stentor* is explored to determine how domestication has happened at a fine scale → Diverged during the Mesoproterozoic Era and belong to the class Heterotrichea
- PiggyBacs are hypothesized to be homologs of the same gene family.



Methodology

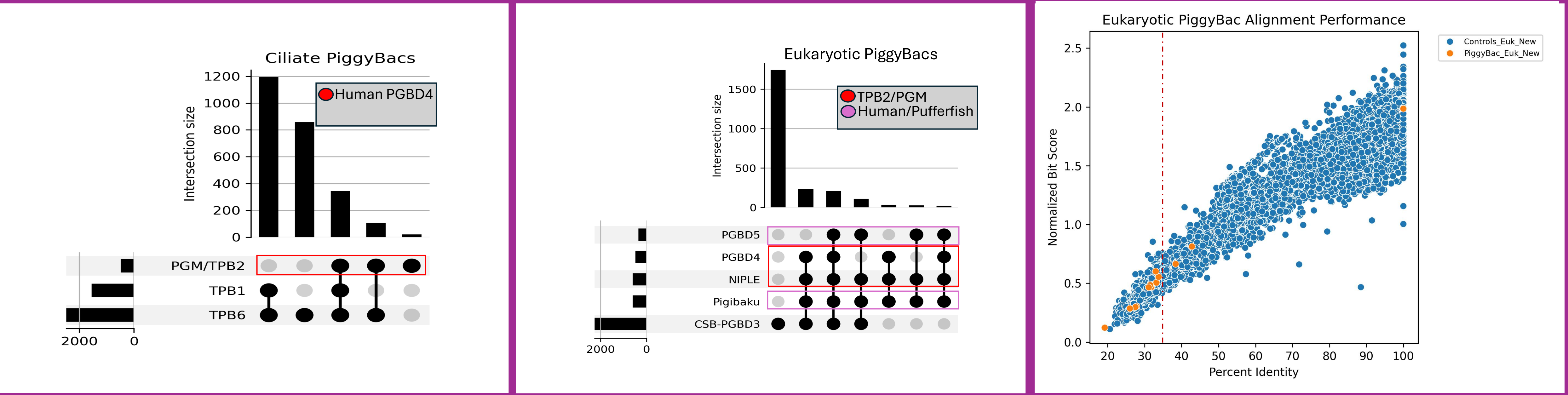


Gene Family Presence



Results

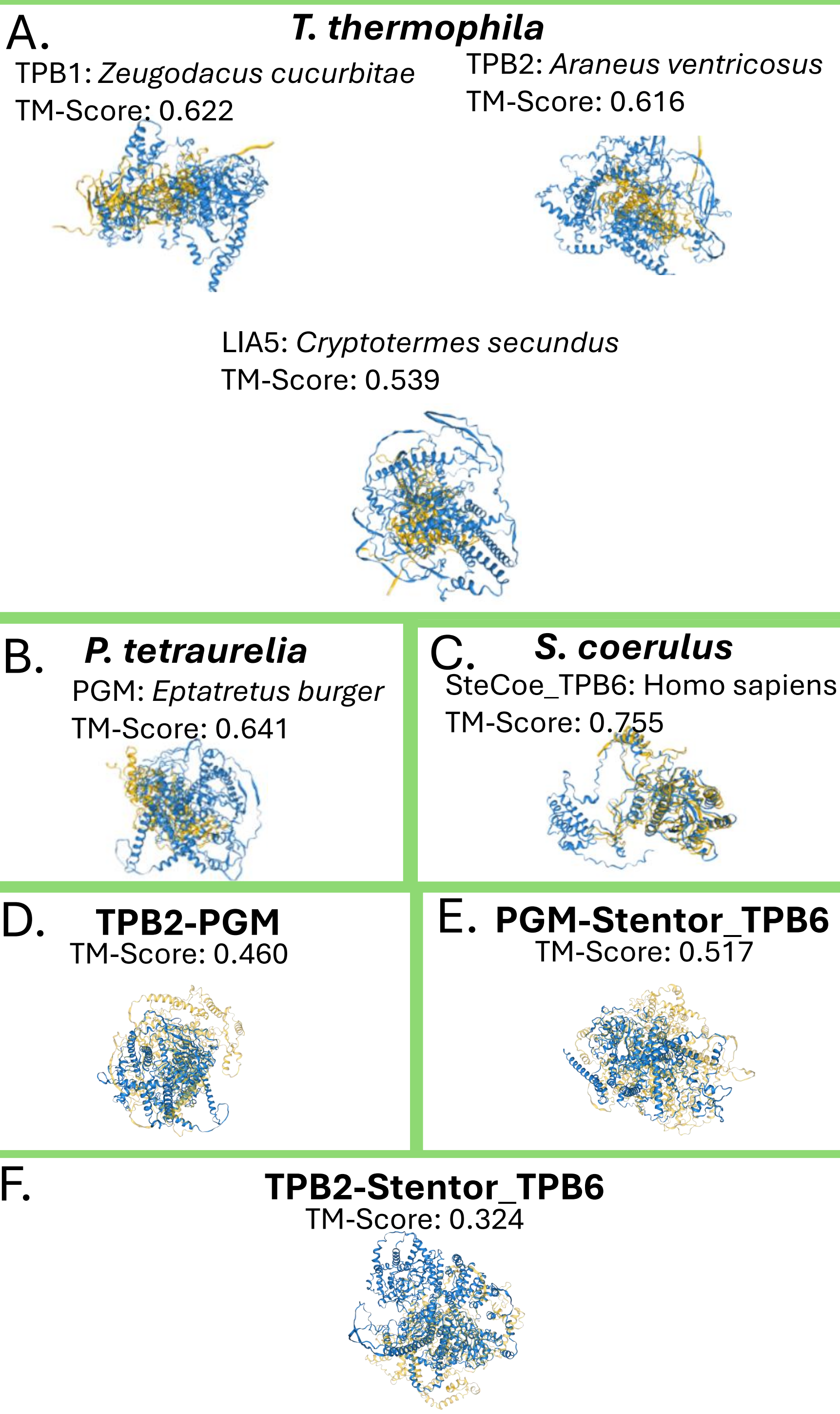
Sequence Homology: Classic Approach



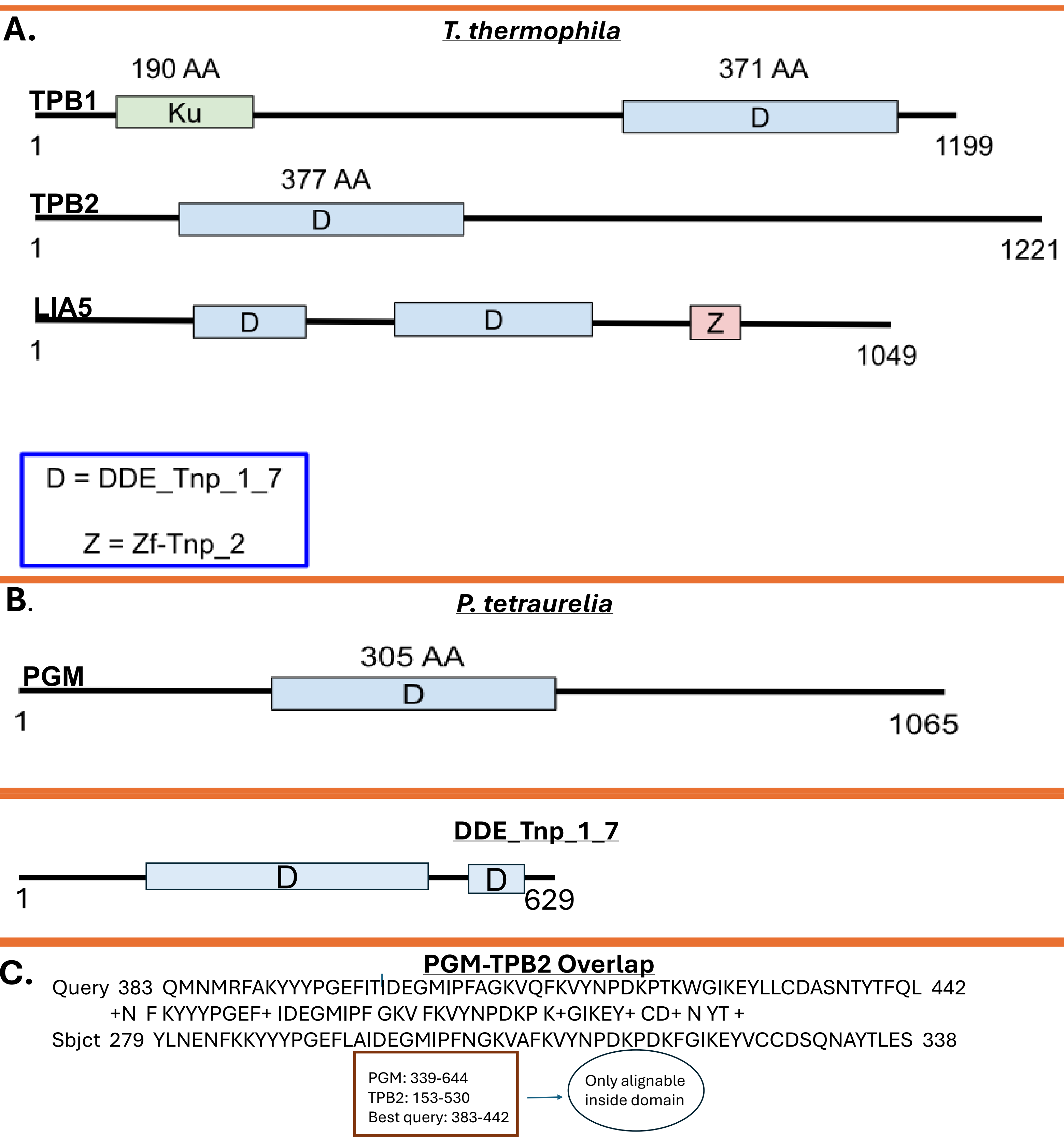
A. Ciliate PiggyBacs are Present in Several Hundred Eukaryotes. TPBs and PGM are in the same group as one of the human PiggyBacs, PGBD4, which stemmed from the Cabbage Moth transposase. They are present in ~1200 eukaryotic taxa. **B.** TPB2 and PGM are in the same group as the human PiggyBac, with a presence in over 500 eukaryotic taxa. While TPB2 and TPB6 are in separate gene families, they are present in a majority of the ~3000 eukaryotes, with tetrahymena PiggyBacs comprising a majority presence.

PiggyBac Sequence Alignments Have Low Matches Against Eukaryotic Sequences. In orange, most PiggyBacs have at most less than 35% percent identity and a low normalized bit score compared to the control eukaryotic housekeeping genes in blue

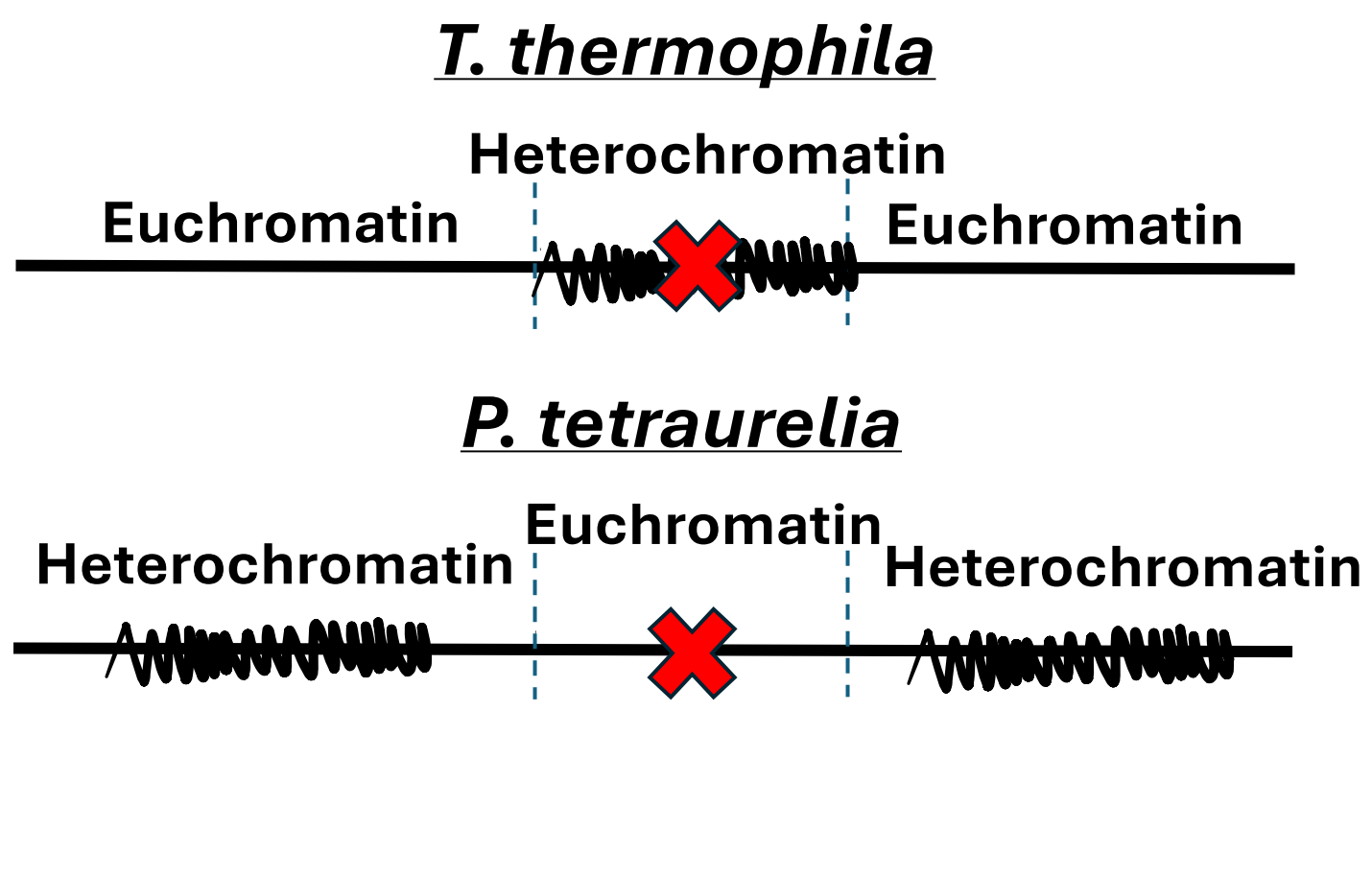
Structural Homology



Domain Homology



Molecular Mechanism



Conclusion

- Previously hypothesized PiggyBacs to be homologs, but based on structural evidence and alignments, their relationship is more complex.
- PiggyBac relationships are complex because they are either fast evolving due to the domain being the only component that is aligned or nonhomologous with multiple origins.
- Due to structural differences within the domain, the accessory parts may be what makes PiggyBac a true excisase.
- Paramecium and Tetrahymena excises recognize elimination sequences differently: heterochromatin v. euchromatin.
- Next steps: Define a genomic toolkit from transposase data to analyze inherent gene editing.

References

Bischerour, J., Bhullar, S., Wilkes, C. D., Régner, V., Mathy, N., Dubois, E., Singh, A., Swart, E., Arnaiz, O., Sperling, L., Nowacki, M., & Bétermier, M. (2018, September 18). Six domesticated PiggyBac transposases together carry out programmed DNA elimination in *Paramecium*. *eLife Sciences Publications Limited*. eLife. <https://doi.org/10.7554/eLife.37927>
Boualligues, M., Rouault, J.-D., Hua-Van, A., Makni, M., & Capi, P. (2017). Molecular Evolution of piggyBac Superfamily: From Selfishness to Domestication. *Genome Biology and Evolution*, 9(2), 323–339. <https://doi.org/10.1093/gbe/evw292>
Dubois, E., Bischerour, J., Marmignon, A., Mathy, N., Régner, V., & Bétermier, M. (2012). Transposon Invasion of the *Paramecium* Germline Genome Countered by a Domesticated PiggyBac Transposase and the NHEJ Pathway. *International Journal of Evolutionary Biology*, 2012(1), 436196. <https://doi.org/10.1155/2012/436196>
Lok, J. (2013). piggyBac: Mobile Genetic Elements. 3(2), e24417. <https://doi.org/10.4161/mge.24417>
Vort, A., Goldman, A. D., Mochizuki, K., & Landweber, L. F. (2013). Transposon Domestication versus Mutualism in Ciliate Genome Rearrangements. *PLoS Genetics*, 9(8), e1003659.

Acknowledgements

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